

Recall: Weil divisor and Cartier divisor

Weil divisor (integral locally Noetherian scheme). $\sum n_z \cdot Z$

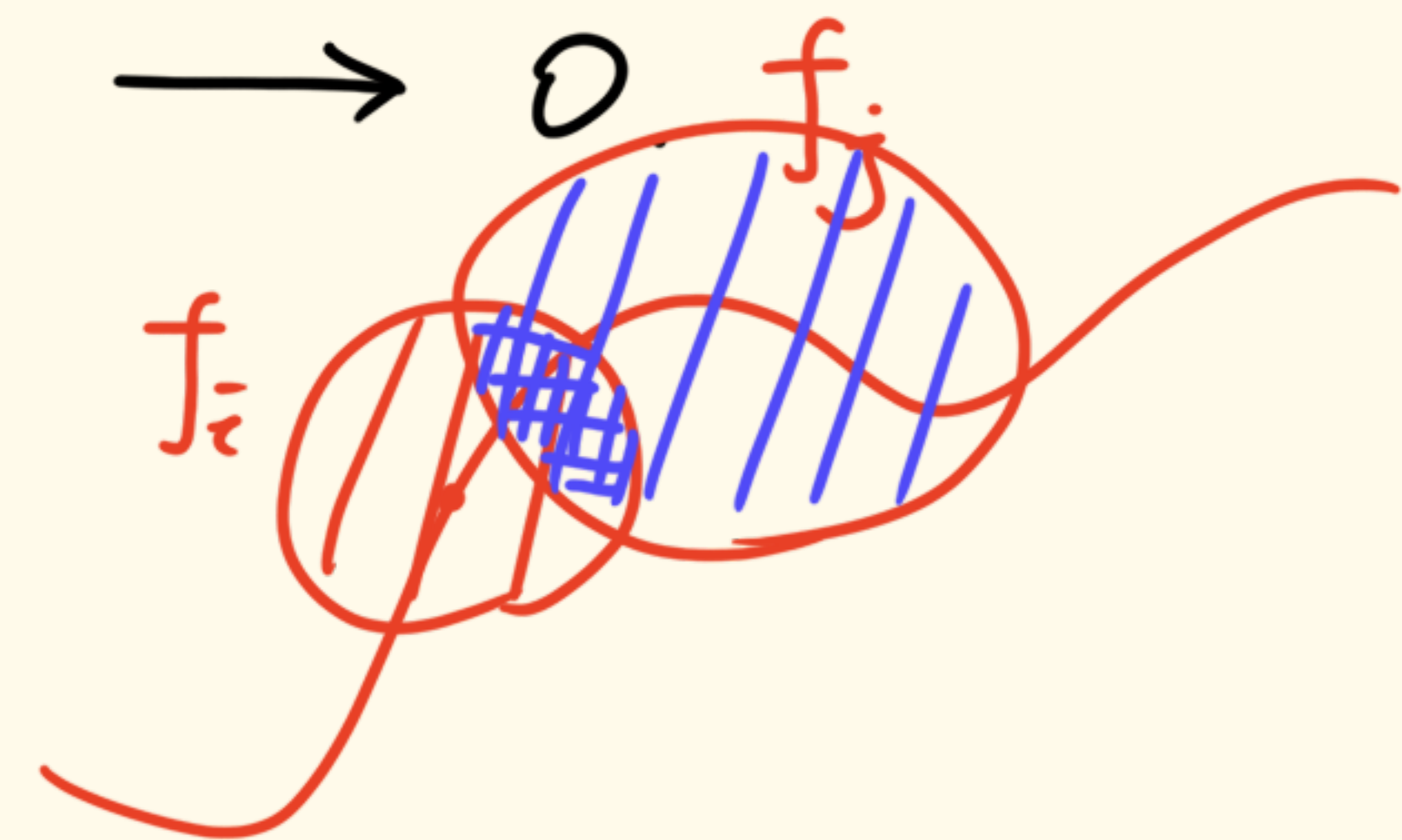
Z integral closed subscheme of codim 1. $\{n_z \neq 0\} \checkmark$

integral scheme: $\exists$ open affine covering $U_i = \text{Spec } A_i$. A_i integral.

Cartier divisor (X is scheme)

$$0 \rightarrow \mathcal{O}_X^\times \rightarrow \mathcal{M}_X^\times \rightarrow \mathcal{M}_X^\times / \mathcal{O}_X^\times \rightarrow 0$$

$\uparrow$ $\uparrow$
rational functions



Hartshorne 141

Def. 1. global section. 2. open cover $\{U_i\}$. $f_i \in \mathcal{M}_X^\times(U_i)$
 $f_i/f_j \in \mathcal{O}_X^\times(U_i \cap U_j)$.

A carrier divisor is principal if it is in the image of the map $H^0(X, \mathcal{M}_x^*) \rightarrow H^0(X, \mathcal{M}_x^* / \mathcal{O}_x^*)$.

right derived functor of global section.

$$H^0(X, \mathcal{M}_x^*) \xrightarrow{f} H^0(X, \mathcal{M}_x^* / \mathcal{O}_x^*) \xrightarrow{g} H^1(X, \mathcal{O}_x^*) \rightarrow 0$$

$$\text{im } f = \text{ker } g.$$

$$\text{Pic}(X).$$

(line bundles up to isomorphism).

cohomology of sheaves. ✓ topic

$$H^0(X, \mathcal{F}) = \mathcal{F}(X). \quad \Gamma \text{ is not exact functor.}$$

$$0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0. \quad 0 \rightarrow \Gamma(X, \mathcal{F}_1) \rightarrow \Gamma(X, \mathcal{F}_2) \rightarrow$$

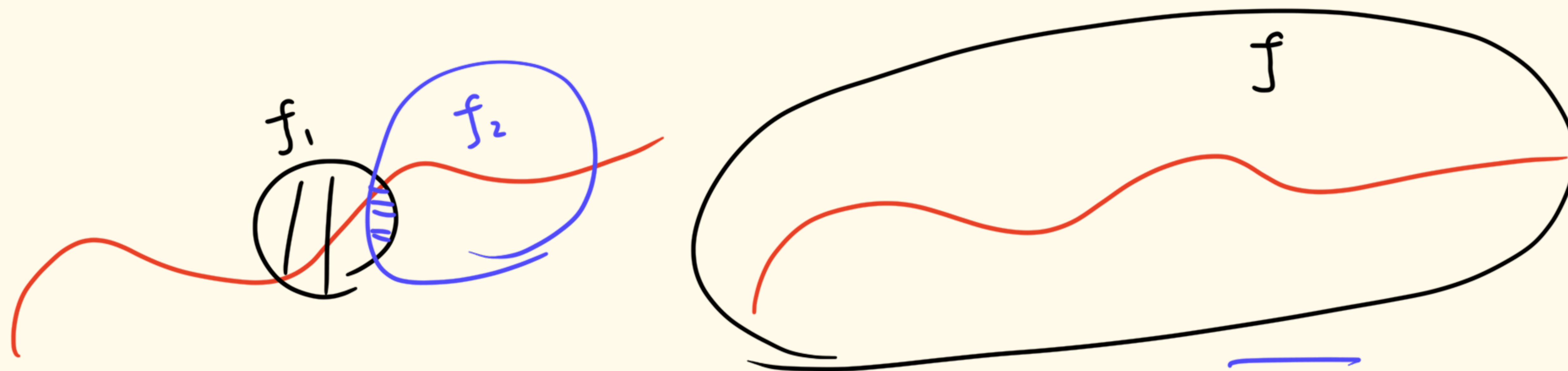
$$\Gamma(X, \mathcal{F}_3) \rightarrow R\Gamma^1(X, \mathcal{F}_1) \rightarrow R\Gamma^1(X, \mathcal{F}_2) \rightarrow \dots$$

↑
right derived functor

Čech cohomology (compute, complex) explicit construction.

Ch 3.1, 3.2

Two Cartier divisors are called linearly equivalent if their difference is principal. $D_1 \sim D_2$ if $\text{Pic}(X) \checkmark$



Weil. "meromorphic function" f , $D = \sum_{P \in X} \text{ord}_P(f) \cdot P$.

D is principal.

A Cartier divisor is effective if it can be represented by $\{(U_i, f_i)\}$ where $f_i \in \mathcal{O}_X(U_i) - \{0\}$ ^{-{0}} regular element.

Weil divisor. $\sum_Z n_Z \cdot Z, n_Z \geq 0. \{n_Z \neq 0\}.$

only zeros, no poles $\frac{(z_1-1)(z_2-2)}{z_3-3} \in \mathcal{M}_X \quad z_1 z_2 - z_1 - 2z_1 + 3$

D effective can be represented closed subscheme.



$V(I)$
 $\downarrow$
ideal sheaf.

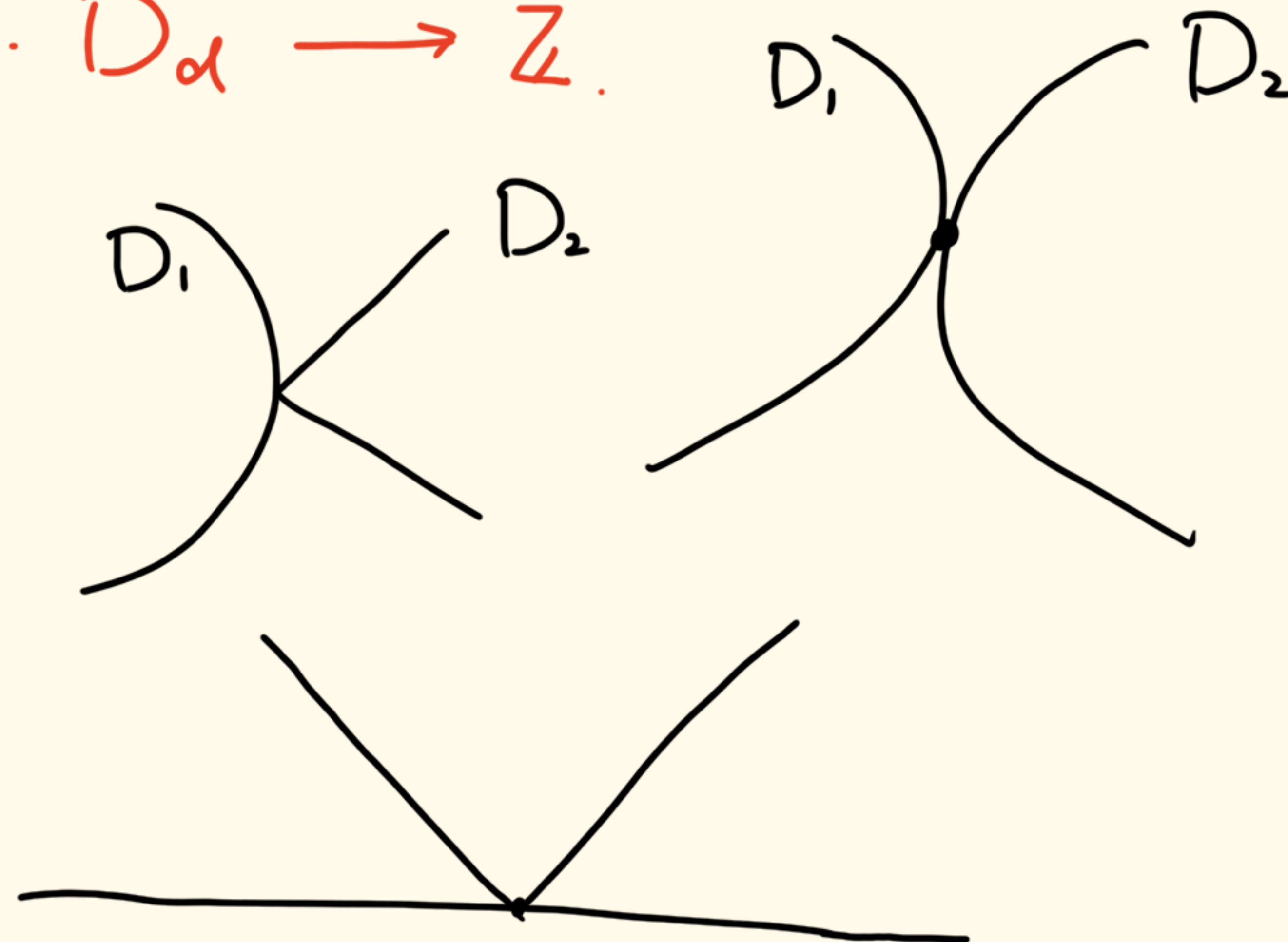
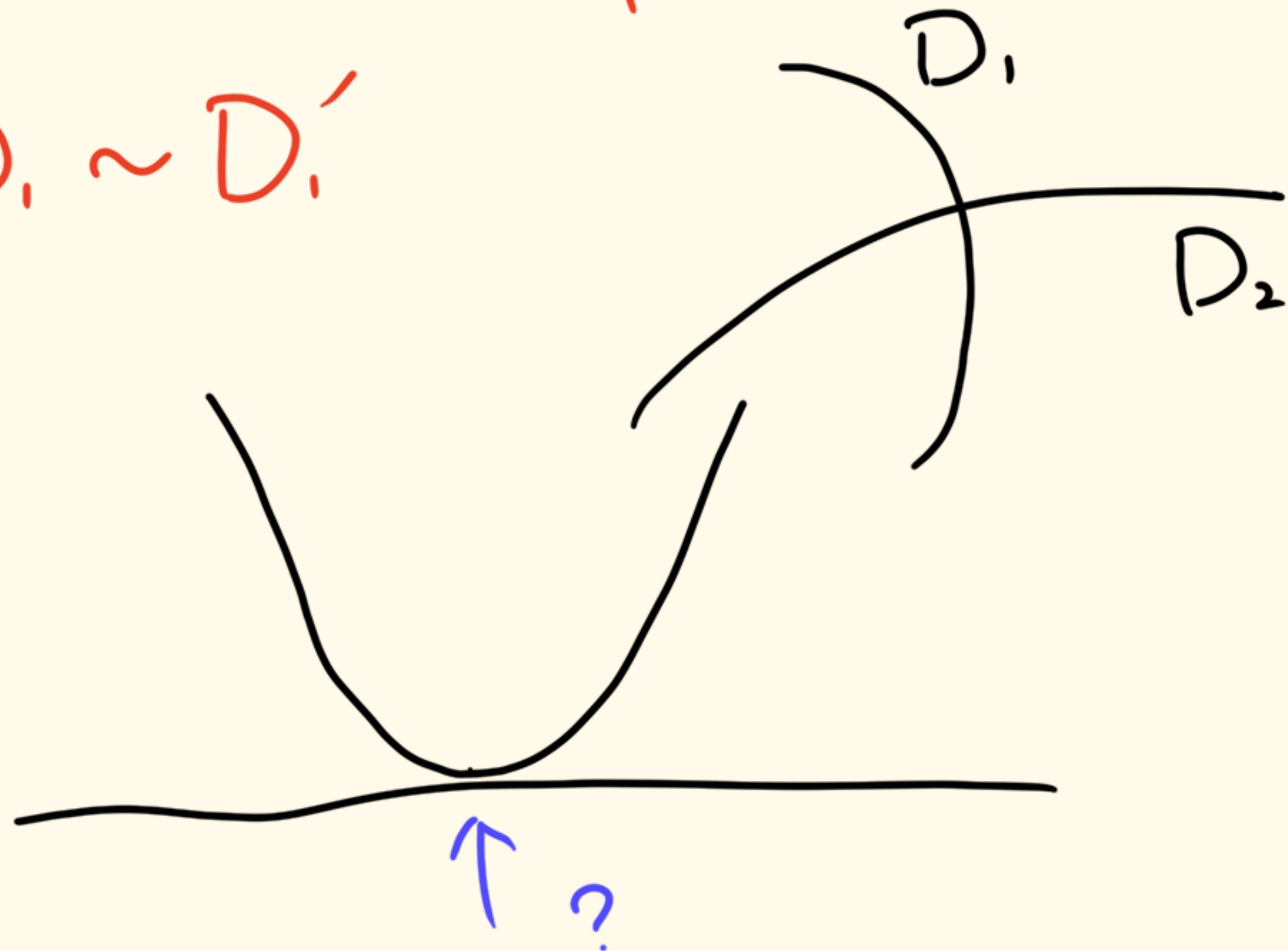
4.2. **Projective varieties and Lorentzian polynomials.** Let Y be a d -dimensional irreducible projective variety over an algebraically closed field $\mathbb{F}$. If $D_1, \dots, D_d$ are Cartier divisors on Y , the intersection product $(D_1 \cdot \dots \cdot D_d)$ is an integer defined by the following properties:

- the product $(D_1 \cdot \dots \cdot D_d)$ is symmetric and multilinear as a function of its arguments,
- the product $(D_1 \cdot \dots \cdot D_d)$ depends only on the linear equivalence classes of the D_i , and
- if $D_1, \dots, D_n$ are effective divisors meeting transversely at smooth points of Y , then

$$(D_1 \cdot \dots \cdot D_d) = \#D_1 \cap \dots \cap D_d. \quad \underline{\text{count}}$$

Intersection product $D_1 \cdot \dots \cdot D_d \rightarrow \mathbb{Z}$.

$D_1 \sim D_1'$



$$\dim \left(\bigcap_{i=1}^d T_p(D_i) \right) = \dim Y - \sum_{i=1}^d \operatorname{codim}(D_i) = 0.$$

$$D \text{ is Cartier divisor} \xleftrightarrow{D \mapsto \mathcal{L}(D)} \operatorname{Pic}(X)$$

Define $\mathcal{L}(D) \subseteq \mathcal{M}_X^\times$ subsheaf generated by f_i^{-1} on U_i .

$D = \sum (U_i, f_i)$. $f_i \in \mathcal{O}_X(U_i) - \{\text{zero divisors}\}$. regular element.

$$\mathcal{L}(D)|_{U_i} \cong f_i^{-1} \cdot \mathcal{O}_X|_{U_i} \quad \xrightarrow{\text{sheafification}} \quad \boxed{\mathcal{L}(D) \text{ invertible sheaf}}$$

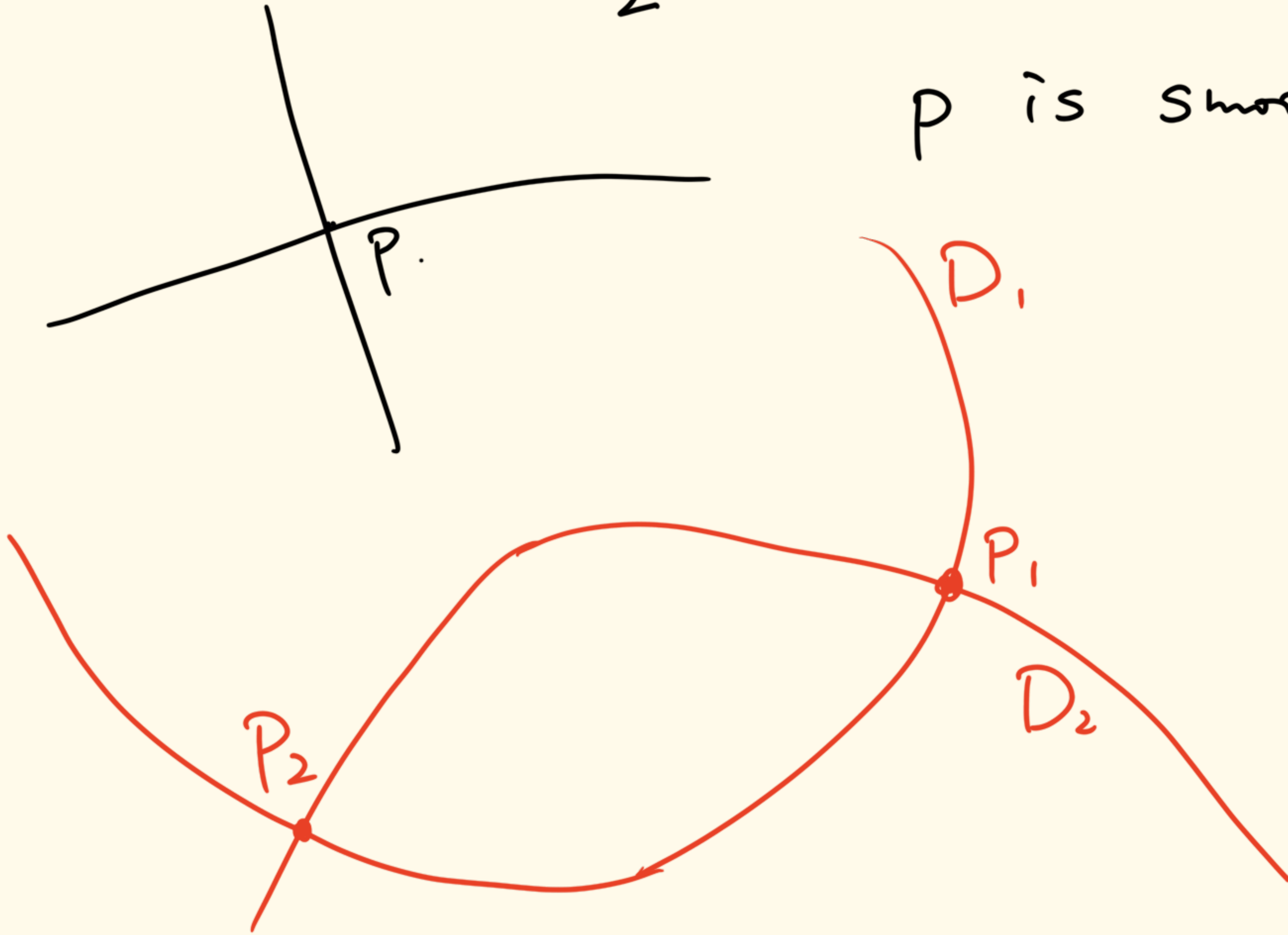
If D is effective, $\mathcal{L}(D)$ is an ideal sheaf of $\mathcal{O}_X$.

$$0 \rightarrow \mathcal{L}(D) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_D \rightarrow 0$$

$m_P \subseteq \mathcal{O}_{D,P}$. $T_P(D) = (m_P / m_P^2)^*$ tangent space.

$\mathcal{O}_{D,P} \subseteq \mathcal{O}_{X,P}$. $\bigcap_{i=1}^d T_P(D) = 0$. Linear spaces.

P is smooth on each D_i .



$D = D_1 - D_2$. both effective

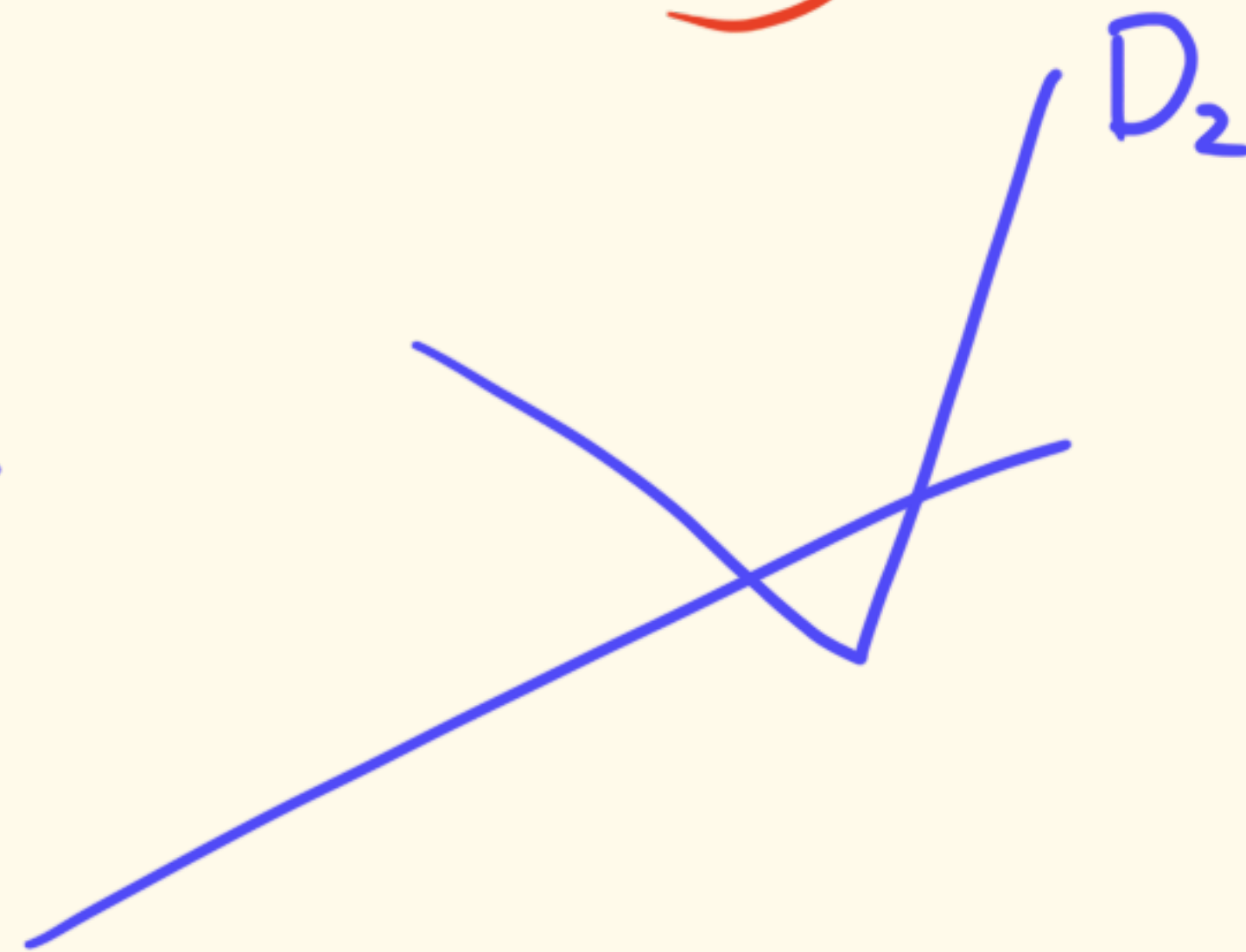
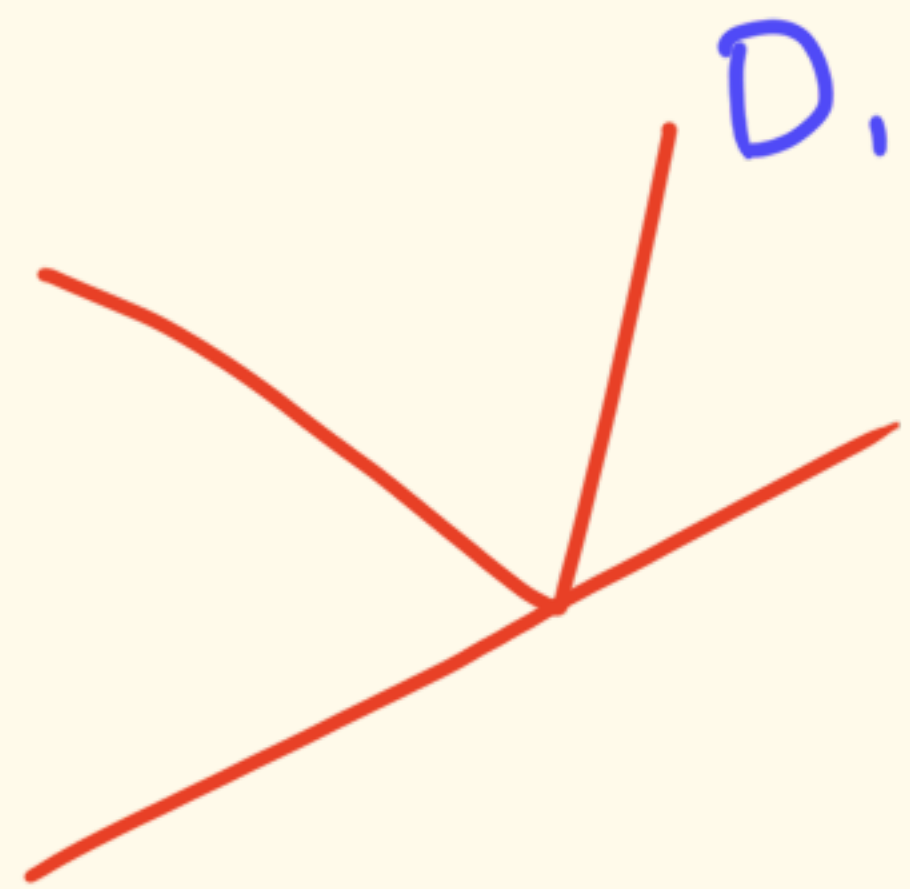
multilinear $\Rightarrow$ expand

equivalence

Chow's moving lemma

$(D_1, \dots, D_d)$ $\downarrow D_d'$

case 3.



$(12H) \checkmark$

$$H = \frac{1}{3} D_1 + \frac{9}{4} D_2$$

在 $\mathbb{R}$ 上没有意义.

volume polynomial.

$$H = (H_1, \dots, H_n)$$

$\mathbb{Q}$ -divisor on Y

$$w = (w_1, \dots, w_n)$$

$$\text{vol}_H(w) := (w_1 H_1 + \dots + w_n H_n)^d = \sum_{\alpha \in \Delta_n^d} \frac{d!}{\alpha!} V_\alpha(H) w^\alpha$$

intersection product over $\mathbb{Q}$.

$$V_\alpha(H) := (\underbrace{H_1 \cdots H_1}_{\alpha_1} \cdots \underbrace{H_n \cdots H_n}_{\alpha_n})$$

在 $\mathbb{R}$ 上? ???

volume polynomial.

Thm. $\text{vol}_H(\omega)$ is a Lorentzian polynomial if $H_1 \cdots H_n$ are nef divisors on Y .

A $\mathbb{Q}$ -divisor D is nef if $(D, C) \geq 0 \quad \forall$ irreducible curve C .

$\mathbb{P}^2$

D

C

$$(D, C) = \deg(D) \cdot \deg(C) \rightarrow \deg(D) \geq 0.$$

nef ~~is~~ effective.

Thm.

Every nef divisor is a limit of ample divisors.

Hartshorne Chapter 2.7.